

Decent Coaching Classes
VIII SB
Mind Your Math - IV
Answer Key

Q.1 (A) Choose the correct alternative. (4)

1. (B) orthocentre
2. (C) $a^3 - 3a^2b + 3ab^2 - b^3$
3. (A) $(a + b)(a^2 - ab + b^2)$
4. (C) xy

Q.1 (B) (any ONE) (1)

1. The orthocentre of a right-angled triangle lies at the vertex of the right angle.
2. If x varies inversely as y , $x \propto 1/y$, so the equation of variation is $x = k/y$ (k is the constant of variation).

Q.2 (any FOUR, 2 marks each) (8)

1. Expand: $(2x - 3y)^3$

$(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$; here $a = 2x$, $b = 3y$

$$\begin{aligned}(2x - 3y)^3 &= (2x)^3 - 3(2x)^2(3y) + 3(2x)(3y)^2 - (3y)^3 \\ &= 8x^3 - 36x^2y + 54xy^2 - 27y^3\end{aligned}$$

2. Factorise: $3x^2 + 14x + 15$

Product = $3 \times 15 = 45$; factors of 45 with sum 14: 9 and 5

$$\begin{aligned}3x^2 + 9x + 5x + 15 &= 3x(x + 3) + 5(x + 3) \\ &= (x + 3)(3x + 5)\end{aligned}$$

3. Simplify: $(x^2 - 9y^2) \div (x - 3y)$

$$(x^2 - 9y^2) = (x - 3y)(x + 3y)$$

$$\therefore (x - 3y)(x + 3y) \div (x - 3y) = x + 3y$$

4. m varies directly as $n \Rightarrow m = kn$

$$154 = k \times 7 \Rightarrow k = 22$$

$$\text{When } n = 14, m = 22 \times 14 = 308$$

5. Diagram question

Since $\text{seg } PX \perp QR$, $\text{seg } PX$ is the altitude of $\triangle PQR$ from vertex P .

Since Y is the midpoint of QR , $\text{seg } PY$ is the median of $\triangle PQR$ from vertex P .

Q.3 (any THREE, 3 marks each) (9)

1. Construction

Steps: (i) Draw an obtuse-angled $\triangle STV$. (ii) Find the midpoints of the three sides. (iii) Join each vertex to the midpoint of the opposite side to get the three medians. (iv) Mark their point of concurrence as the centroid G .

Note: The centroid of any triangle (acute, right or obtuse) always lies in the interior of the triangle.

2. Diagram question

The centroid divides each median in the ratio 2 : 1 from the vertex.

$$AG : GP = 2 : 1 \text{ and } l(AG) = 8.4 \text{ cm} \Rightarrow l(GP) = 8.4 \div 2 = 4.2 \text{ cm}$$

$$BG : GQ = 2 : 1 \text{ and } l(BQ) = l(BG) + l(GQ) = 9 \text{ cm} \Rightarrow l(BG) = (2/3) \times 9 = 6 \text{ cm}$$

$$\therefore l(GP) = 4.2 \text{ cm and } l(BG) = 6 \text{ cm}$$

3. Expand: $(2p + 3q + 4)^2$

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ac; a = 2p, b = 3q, c = 4$$

$$= (2p)^2 + (3q)^2 + 4^2 + 2(2p)(3q) + 2(3q)(4) + 2(2p)(4)$$

$$= 4p^2 + 9q^2 + 16 + 12pq + 24q + 16p$$

4. Factorise: $3p^2 - 3p - 60$

$$= 3(p^2 - p - 20)$$

Factors of -20 with sum -1 : -5 and 4

$$= 3(p - 5)(p + 4)$$

5. Simplify: $(3x + 2y)^3 - (3x - 2y)^3$

Using $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$, with $a = 3x + 2y$, $b = 3x - 2y$:

$$a - b = 4y$$

$$a^2 + ab + b^2 = (9x^2 + 12xy + 4y^2) + (9x^2 - 4y^2) + (9x^2 - 12xy + 4y^2) = 27x^2 + 4y^2$$

$$\therefore (3x + 2y)^3 - (3x - 2y)^3 = 4y(27x^2 + 4y^2) = 108x^2y + 16y^3$$

Q.4 (any TWO, 4 marks each)**(8)**

1. (a) The orthocentre O of a right-angled triangle lies at the vertex where the right angle is formed — because the two sides containing the right angle are themselves altitudes (each perpendicular to the other), so they (and the third altitude) meet exactly at that vertex.

(b) For an equilateral triangle, the centroid G lies in the interior of the triangle. In fact, for an equilateral triangle, the centroid, circumcentre, incentre and orthocentre all coincide at the same point.

2. Inverse variation: number of workers $\times$ number of hours = constant (k)

(a) $k = 18 \times 40 = 720$; equation of variation: $n \times h = 720$ (or $n = 720/h$)

(b) When $h = 24$: $n \times 24 = 720 \Rightarrow n = 30$

$\therefore$ 30 workers will be required.

3. (a) Factorise: $27p^3 + 8q^3$

$$= (3p)^3 + (2q)^3 = (3p + 2q)[(3p)^2 - (3p)(2q) + (2q)^2]$$

$$= (3p + 2q)(9p^2 - 6pq + 4q^2)$$

(b) Simplify: $(a^2 - 49) \div (a^2 + 3a - 28)$

$$a^2 - 49 = (a - 7)(a + 7)$$

$a^2 + 3a - 28$: factors of -28 with sum 3 are 7 and $-4 \Rightarrow (a + 7)(a - 4)$

$$\therefore (a - 7)(a + 7) \div [(a + 7)(a - 4)] = (a - 7)/(a - 4)$$

— *End of Answer Key* —